

1. BUSINESS SCENARIO

A Smart Food Delivery Management System is designed to connect customers, restaurants, and delivery drivers through a centralized platform. Customers can browse restaurant menus, place orders, and make payments, while restaurants prepare meals and drivers deliver them. The system ensures efficient coordination between all parties and maintains accurate records of all transactions.

The system relies on a relational database to manage structured data such as customers, orders, menu items, and deliveries. It ensures data integrity, supports concurrent transactions (e.g., multiple customers placing orders simultaneously), and automates business processes like updating order status and assigning drivers. Additionally, it enables analytics such as tracking top-performing restaurants and delivery efficiency.

2. ENTITIES WITH ATTRIBUTES (WITH DATA TYPES)

Customer

Attribute	Data Type	Description
customer_id	INTEGER (PK)	Unique customer ID
name	VARCHAR(20)	Customer name (<i>Text ≤ 20 chars</i>)
phone	VARCHAR(15)	Contact number
address	VARCHAR(255)	Address (<i>Text ≤ 255 chars</i>)
created_at	DATE	Account creation date
status	CHAR(1)	('A','T') (<i>Single char limited</i>)

Restaurant

Attribute	Data Type	Description
restaurant_id	INTEGER (PK)	Unique restaurant ID
name	VARCHAR(20)	Restaurant name (<i>Text ≤ 20 chars</i>)
cuisine_type	VARCHAR(20)	('Italian','Asian','FastFood') (<i>Limited text choices</i>)
opening_time	DATE	Opening time
closing_time	DATE	Closing time
status	CHAR(1)	('A','T') (<i>Single char limited</i>)

Menu_Item

Attribute	Data Type	Description
item_id	INTEGER (PK)	Unique item ID
restaurant_id	INTEGER (FK)	References Restaurant (<i>Foreign Key</i>)
name	VARCHAR(20)	Item name (<i>Text ≤ 20 chars</i>)
price	DECIMAL(8,2)	Item price (<i>Decimal</i>)
category	VARCHAR(20)	('Meal','Drink','Dessert') (<i>Limited text</i>)
is_available	CHAR(1)	('Y','N') (<i>Single char limited</i>)

Order

Attribute	Data Type	Description
order_id	INTEGER (PK)	Unique order ID
customer_id	INTEGER (FK)	References Customer
restaurant_id	INTEGER (FK)	References Restaurant
order_time	DATE	Order timestamp (<i>Date</i>)
status	VARCHAR(20)	('Pending','Preparing','Delivered','Cancelled') (<i>Limited text</i>)
total_amount	DECIMAL(10,2)	Total order cost (<i>Decimal</i>)

Order_Item

Attribute	Data Type	Description
order_id	INTEGER (FK, PK)	References Orders
item_id	INTEGER (FK, PK)	References Menu_Item
quantity	INTEGER	<i>Quantity ordered</i>
unit_price	DECIMAL(8,2)	<i>Price per unit</i>
subtotal	DECIMAL(10,2)	Calculated subtotal

Driver

Attribute	Data Type	Description
driver_id	INTEGER (PK)	Unique driver ID
name	VARCHAR(20)	Driver name (<i>Text ≤ 20 chars</i>)
status	CHAR(1)	('A','B') (<i>Single char limited</i>)
total_deliveries	INTEGER	Total deliveries (<i>Integer</i>)
avg_rating	DECIMAL(3,2)	Driver rating (<i>Decimal</i>)

Delivery

Attribute	Data Type	Description
delivery_id	INTEGER (PK)	Unique delivery ID
order_id	INTEGER (FK)	References Orders
driver_id	INTEGER (FK)	References Driver
pickup_time	DATE	<i>Pickup time (Date + Time)</i>
delivery_time	DATE	<i>Delivery time (Date + Time)</i>
delivery_status	VARCHAR(20)	('Assigned','Delivered')

Payment

Attribute	Data Type	Description
payment_id	INTEGER (PK)	Unique payment ID
order_id	INTEGER (FK)	References Orders
amount	DECIMAL(10,2)	<i>Payment amount</i>
method	VARCHAR(20)	<i>('Card','Cash')</i>
status	CHAR(1)	<i>('Y','N')</i>
payment_time	DATE	<i>Payment timestamp (Date + Time)</i>

Review

Attribute	Data Type	Description
review_id	INTEGER (PK)	Unique review ID
order_id	INTEGER (FK)	References Order
rating	INTEGER	Rating (<i>Integer</i>)
comment	VARCHAR(255)	Review comment (<i>Text ≤ 255 chars</i>)

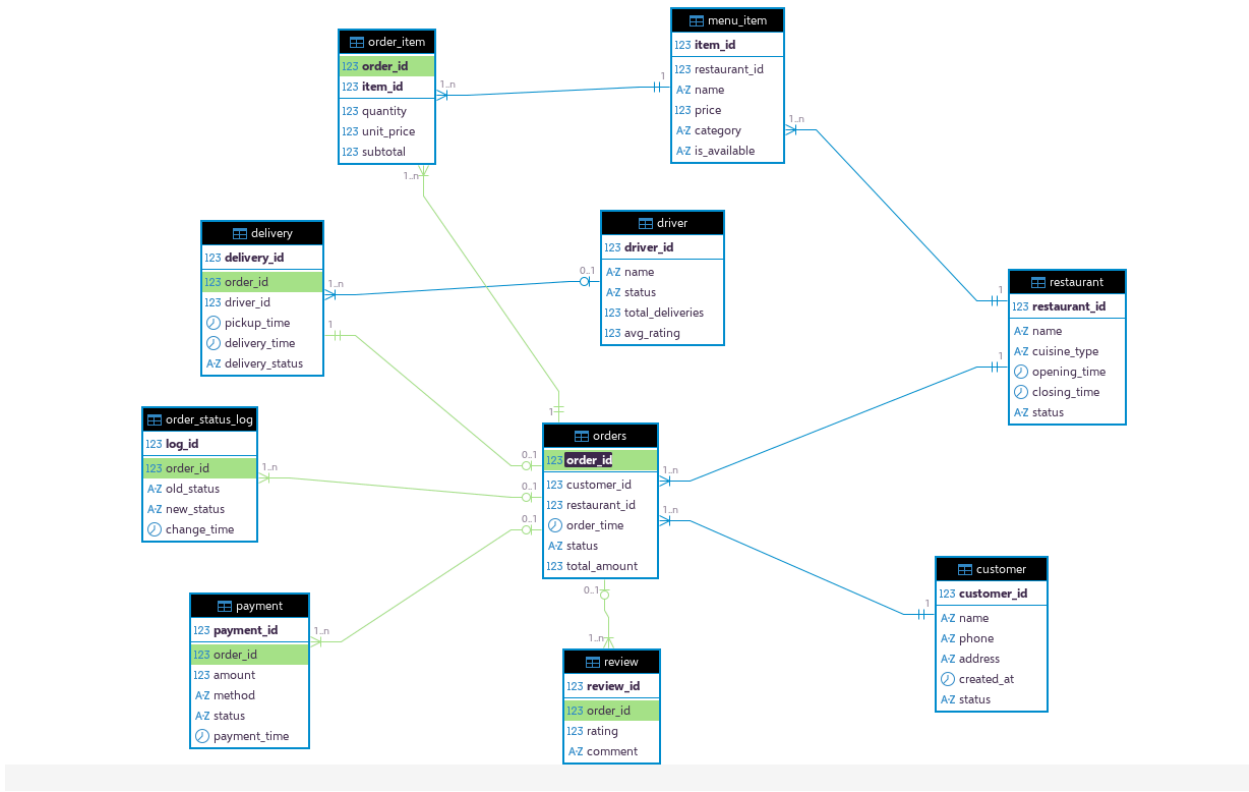
Order_Status_Log

Attribute	Data Type	Description
log_id	INTEGER (PK)	Unique log ID
order_id	INTEGER (FK)	References Orders
old_status	VARCHAR(20)	Previous status
new_status	VARCHAR(20)	New status
change_time	DATE	Time of change (<i>Date + Time</i>)

3. ATTRIBUTE TYPE COVERAGE

Requirement	Where Used
Primary Keys	All entities (customer_id, order_id, etc.)
Foreign Keys	order.customer_id, order_item.item_id, etc.

Integer	IDs, quantity, rating
Decimal	price, total_amount
Date	order_time, payment_time
Limited Text	status, category, method
VARCHAR(20)	names, categories
VARCHAR(255)	address, comment
Single Char	status ('A','T'), ('Y','N')



FOOD DELIVERY DATABASE (FINAL DESIGN)

— SYSTEM

└─ Data Dictionary

- └─ Users, Roles, Privileges
- └─ Metadata for tables, indexes, views
- └─ Stored procedures, triggers metadata

— DATA (HIGH ACTIVITY TABLES 🔥)

- └─ Orders
- └─ Order_Item
- └─ Delivery
- └─ Payment
- └─ Order_Status_Log

— DATA_2 (LOW / MEDIUM ACTIVITY TABLES)

- └─ Customer
- └─ Restaurant
- └─ Menu_Item
- └─ Driver
- └─ Review

— INDEXES (HIGH USAGE INDEXES 🔥)

- └─ idx_orders_customer
- └─ idx_orders_restaurant
- └─ idx_delivery_driver
- └─ idx_payment_order
- └─ idx_order_item_order

— INDEXES_2 (LOW USAGE INDEXES)

- └─ idx_customer_name
- └─ idx_restaurant_name
- └─ idx_menu_item_category



- └─ UNDO
 - └─ Undo segments
 - └─ (Rollback, concurrency, read consistency)
- └─ UNDO_2 (OPTIONAL ADVANCED)
 - └─ Large bulk operations (if implemented)
- └─ TEMP
 - └─ Temporary segments
 - └─ Sorting (ORDER BY)
 - └─ GROUP BY operations
 - └─ Window functions (RANK, SUM OVER)
- └─ TEMP_USER
 - └─ Heavy analytics queries (optional advanced)
- └─ TOOLS
 - └─ Oracle / admin tools
- └─ TOOLS_I
 - └─ Indexes for tools
- └─ USERS
 - └─ User-created objects
 - └─ Views (customer_orders_view, driver_view)
 - └─ Synonyms (orders_syn, delivery_syn)
 - └─ Stored Procedures (place_order, assign_driver)
 - └─ Functions (calculate_total)



Weights & Percentages

Tablespace	Weight	Percent of Total
DATA	100	28%

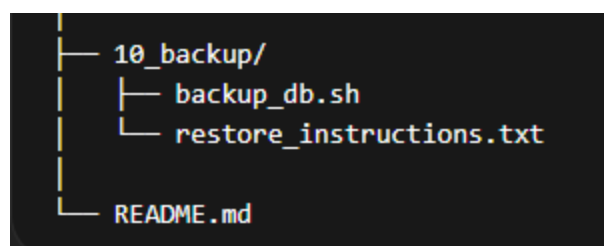
DATA_2	60	17%
INDEXES	80	22%
INDEXES_2	30	8%
UNDO	70	19%
TEMP	40	11%
SYSTEM	30	8%
USERS	50	14%
Total	360	100%

Tablespace Distribution Across Disks

Disk Number	Weight	Contents (Tablespaces)
1	-	Oracle Software
2	100	DATA
3	150	INDEXES, UNDO
4	100	DATA_2, TEMP
5	110	USERS, INDEXES_2, SYSTEM

- 01_db_setup/
 - |— dbca_scripts/ ← (auto-generated by DBCA)
 - |— init_params.sql
 - |— create_tablespace.sql
- 02_schema/
 - |— create_tables.sql
 - |— add_constraints.sql
 - |— create_indexes.sql
- 03_security/
 - |— create_roles.sql
 - |— create_users.sql
 - |— grant_privileges.sql
- 04_business_logic/
 - |— procedures.sql
 - |— functions.sql
 - |— triggers.sql
- 05_views/
 - |— row_level_views.sql
 - |— column_level_views.sql
- 06_data/
 - |— sample_data.csv
 - |— load_data.sql
- 07_transactions/
 - |— concurrency_demo.sql
 - |— isolation_tests.sql
- 08_analysis/
 - |— window_functions.sql
 - |— reporting_queries.sql
 - |— performance_analysis.sql
- 09_admin_scripts/
 - |— startup_db.sh
 - |— shutdown_db.sh
 - |— check_status.sql
 - |— current_user.sql
 - |— active_sessions.sql





Users and accounts

 LAYER 1: PRIVILEGES (LOWEST LEVEL)	
These are actual database actions:	
Table	Privileges
orders	SELECT, INSERT
order_item	SELECT, INSERT
delivery	SELECT, UPDATE
menu_item	SELECT
payment	INSERT, SELECT
customer	SELECT, UPDATE
review	INSERT
	

◆ Application Roles Table

Application Role	Purpose	Privileges	
place_order_role	Customer places order	INSERT orders, SELECT menu_item	
manage_order_items_role	Add items	INSERT order_item	
process_payment_role	Payment	INSERT payment	
deliver_order_role	Driver updates delivery	UPDATE delivery	
update_customer_role	Modify profile	UPDATE customer	
review_order_role	Leave review	INSERT review	
view_menu_role	Browse menu	SELECT menu_item	
admin_full_access_role	Admin operations	ALL privileges	



LAYER 4: USERS (REAL ACCOUNTS)

User	Assigned Role
cust_1	customer_role
cust_2	customer_role
driver_1	driver_role
rest_1	restaurant_role
admin_1	admin_role

<https://chatgpt.com/share/69e13f26-38b8-832c-8aa7-dfb306f79f67>